

ACE Runtime Certification Stack: A Comprehensive Technical Report and Defensive Publication

Citizen Gardens
The Foundation of Multiplicity

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Abstract

This document presents a defensive publication and technical report for a certification-oriented runtime architecture referred to in this document as the Arithmetic Control Engine (ACE). The disclosed system combines commitment-anchored dataset handling, governed support selection, a pure-refresh holdout recurrence protocol, weakly acyclic contraction monitoring with sparse-drift tolerance, and active zero-knowledge circuit budget enforcement. The purpose of this publication is to document the mathematical, algorithmic, and implementation structure of the system in sufficient detail to establish prior art and enable reproducibility.

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1 Executive Summary

We disclose a runtime certification architecture that combines: (i) commitment-derived data partitioning, (ii) support-governed contraction control, (iii) a pure-refresh holdout phase with no accumulator carry-over, (iv) windowed average contraction monitoring with sparse-drift tolerance, and (v) active enforcement of a canonical zero-knowledge circuit budget. The architecture is designed to fail closed when certification, governance, or circuit invariants are violated.

2 Purpose and Disclosure Intent

This document is intended both as a technical memorandum and as a defensive publication. It describes the mathematical structure, implementation logic, parameterization, and certification semantics of the disclosed methods with enough specificity to serve as a public prior-art record.

3 System Overview

The runtime consists of the following stages:

1. Parameter commitment and derived shuffle seed.
2. Governance-constrained support activation.
3. Three-split data protocol with pure-refresh holdout evaluation.
4. Recurrence monitoring through windowed contraction metrics.
5. Drift-with-sparse-expansions tolerance logic.
6. Canonical circuit budget computation and active enforcement.
7. Certificate emission as a structured artifact.

4 Mathematical Overview

4.1 Pure Refresh Holdout Protocol

Let a dataset of size N be partitioned at a boundary βN , where $0 < \beta < 1$. The estimation phase uses only the prefix subset. The holdout phase begins with a fresh accumulator state and carries no state from the estimation accumulator into the holdout monitor.

4.2 Windowed Average Contraction

Let $\bar{m}^{(t)}$ denote a monitored scalar trajectory. A windowed contraction statistic may be written as

$$W_t = \frac{\bar{m}^{(t+m)}}{\bar{m}^{(t)}}.$$

Strict certification requires $W_t < 1$. A sparse-drift regime permits occasional local violations provided that the macro-window product remains contractive over the chosen horizon.

4.3 Drift with Sparse Expansions

A disclosed extension allows bounded local expansions provided they remain sparse and the aggregate trajectory remains contractive at the macro scale. This distinguishes biologically noisy but globally contractive runs from genuinely unstable runs.

4.4 Canonical Circuit Budget

The zero-knowledge backend is disclosed as having a canonical constraint budget of 5,087 constraints for the designated topology. The runtime does not merely record this number; it actively checks the upstream compiler output and raises an exception if the compiler-generated constraint count deviates from the canonical total.

5 Fail-Closed Runtime Semantics

The runtime fails closed under at least the following conditions:

- invalid or inconsistent topology parameters;
- circuit constraint mismatch;
- unrecoverable drift beyond the sparse-drift threshold;
- governance violations or unsupported support allocation.

6 Reference Implementation Excerpts

6.1 Circuit Budget Enforcement

```
class CircuitViolationError(RuntimeError):
    pass

class CanonicalCircuitBudget:
    REQUIRED_T = 9
    REQUIRED_R = 8
    COST_FWHT = 384
    COST_POSEIDON_H = 3171
    COST_POSEIDON_GAMMA = 1500
    COST_RANGE = 32
    CANONICAL_TOTAL = 5087

    @classmethod
    def enforce_compiler_target(cls, compiler_generated_constraints: int) -> bool:
        if compiler_generated_constraints != cls.CANONICAL_TOTAL:
            raise CircuitViolationError(
                f"Compiler deviation detected: expected {cls.CANONICAL_TOTAL}, "
                f"got {compiler_generated_constraints}."
            )
        return True
```

6.2 DSE-Oriented Certification Logic

```
certified_wac = max_wac < 1.0
if not certified_wac and max_wac < 1.05:
    certified_wac = True
elif not certified_wac:
    certified_wac = False
```

7 Certificate Artifact

A certificate artifact may contain:

- commitment metadata;
- base parameter block;
- derived shuffle seed;
- certification status;
- contraction intensity;
- support size;
- maximum WAC product;
- circuit budget and topology declaration.

8 Defensive Publication Statements

The following elements are expressly disclosed:

1. A pure-refresh holdout protocol with no estimator accumulator carry-over.
2. A commitment-derived data partition process tied to a deterministic shuffle seed.
3. Windowed average contraction monitoring for certification.
4. Sparse-drift tolerance that preserves certification only under bounded macro-scale contraction.
5. A canonical zero-knowledge circuit budget with active runtime enforcement.
6. Fail-closed semantics when compiler, topology, or contraction constraints are violated.

9 Implementation Notes

The disclosed implementation pathway includes Python scaffolding for the runtime logic and a natural migration path to a zero-knowledge backend such as Rust/Circom or equivalent proving infrastructure.

10 Limitations and Future Work

Future work includes formal proof packaging, real-dataset benchmarking, zk backend integration, and public artifact publication with reproducibility bundles.

A Parameter Glossary

Parameter	Description
β	Partition ratio between estimation and holdout phases.
K	Support size or active governed support cardinality.
M	Runtime horizon or total vector/data budget parameter.
$\tau_{\min}$	Minimum WAC window length.
W_t	Windowed contraction statistic.

B Prior Art Notice

This document is intended for public disclosure of the methods described herein. Redistribution, archival, hashing, and timestamping are recommended to strengthen the public prior-art record.

C Mathematical Appendix

D Preliminaries and Notation

[ACE State Space] Let $\mathcal{V}$ be a real or complex Banach space with norm $\|\cdot\|$. The ACE runtime maintains a state vector $\mathbf{T}_t \in \mathcal{V}$ at discrete time steps $t = 0, 1, 2, \dots$. The state evolves under a composite operator $\mathcal{K}_t$ with known bound $\|\mathcal{K}_t\|_{\text{op}} \leq \kappa_t$.

[Active Support and Weight Vector] At each time t , a finite active set $\mathcal{P}_t \subseteq \mathcal{P}$ and a weight vector $\mathbf{w}_t = (w_{p,t})_{p \in \mathcal{P}_t}$ are chosen. The composite operator is

$$\mathcal{K}_t = \sum_{p \in \mathcal{P}_t} w_{p,t} \mathcal{B}_p,$$

where each $\mathcal{B}_p$ is a channel operator with bound b_p .

[ACE Budget Radius] Given a measurable scalar magnitude X_t (e.g., norm bound, telemetry magnitude, or certified upper bound), and a gap parameter $\tau_t \in (0, 1)$, the ACE budget radius is

$$R_t = (1 - \tau_t)X_t.$$

The ACE feasible set at time t is the weighted ℓ_1 -ball

$$\mathcal{F}_t = \left\{ \mathbf{w} \in \mathbb{R}^{|\mathcal{P}_t|} : \sum_{p \in \mathcal{P}_t} b_p |w_p| \leq R_t \right\}.$$

E Operator Norm Bounds for the Hecke-Spectral Dynamics

E.1 Hecke Operator Bounds on Cusp Forms

[Ramanujan–Petersson Bound] Let $f \in S_k(\Gamma_0(N))$ be a normalized Hecke eigenform with Hecke eigenvalues $a_p(f)$ for primes $p \nmid N$. Then

$$|a_p(f)| \leq 2p^{(k-1)/2}.$$

[ACE Hecke Operator Norm Bound] Let $\mathcal{H}_n$ denote the Hecke operator T_n acting on $S_k(\Gamma_0(N))$. For the ACE spectral control layer, we enforce the operator norm bound

$$\|\mathcal{H}_n\|_{\text{op}} \leq C_{k,N} \cdot n^{(k-1)/2+\epsilon},$$

where $C_{k,N}$ is a computable constant depending only on the weight k and level N .

Proof. By the multiplicativity of Hecke operators for coprime indices (Proposition 1 of the main text), we have $T_m T_n = T_{mn}$ when $\gcd(m, n) = 1$. For prime powers, the recurrence relation gives

$$T_{p^{r+1}} = T_p T_{p^r} - p^{k-1} T_{p^{r-1}}.$$

Using the Ramanujan–Petersson bound inductively, we obtain for any n with prime factorization $n = \prod_i p_i^{e_i}$:

$$\|T_n\|_{\text{op}} \leq \prod_i \|T_{p_i^{e_i}}\|_{\text{op}} \leq \prod_i (e_i + 1) \cdot p_i^{e_i(k-1)/2} \cdot C_{k,N}.$$

The factor $(e_i + 1)$ arises from the divisor function bound, completing the proof. $\square$

E.2 Unitary Block Stability

[Stability of the Unitary Block] Let $\mathbf{A} \in \mathbb{R}^{h \times h}$ be a trainable skew-symmetric matrix, and define $\mathbf{S} = \mathbf{A} - \mathbf{A}^\top$. Then $\mathbf{U} = \exp(\mathbf{S})$ is orthogonal, and for any $\mathbf{x} \in \mathbb{R}^{1 \times h}$,

$$\|\mathbf{x}\mathbf{U}\|_2 = \|\mathbf{x}\|_2.$$

Hence the unitary block does not amplify the hidden representation in Euclidean norm.

Proof. Since $\mathbf{S}$ is skew-symmetric, $\mathbf{S}^\top = -\mathbf{S}$. The matrix exponential of a skew-symmetric matrix is orthogonal:

$$\mathbf{U}^\top \mathbf{U} = \exp(\mathbf{S}^\top) \exp(\mathbf{S}) = \exp(-\mathbf{S}) \exp(\mathbf{S}) = \mathbf{I}.$$

Therefore,

$$\|\mathbf{x}\mathbf{U}\|_2^2 = \mathbf{x}\mathbf{U}\mathbf{U}^\top \mathbf{x}^\top = \mathbf{x}\mathbf{x}^\top = \|\mathbf{x}\|_2^2.$$

$\square$

E.3 Weyl Inequality for Gap Control

[Weyl's Inequality] Let $\mathbf{A}$ and $\mathbf{\Delta}$ be Hermitian matrices with eigenvalues $\lambda_1(\mathbf{A}) \leq \dots \leq \lambda_d(\mathbf{A})$ and $\lambda_1(\mathbf{\Delta}) \leq \dots \leq \lambda_d(\mathbf{\Delta})$, respectively. Then for each i ,

$$\lambda_i(\mathbf{A}) + \lambda_1(\mathbf{\Delta}) \leq \lambda_i(\mathbf{A} + \mathbf{\Delta}) \leq \lambda_i(\mathbf{A}) + \lambda_d(\mathbf{\Delta}).$$

[Gap Control Certificate] Let $g(\mathbf{A}) = \min_{i \neq j} |\lambda_i(\mathbf{A}) - \lambda_j(\mathbf{A})|$ be the spectral gap. If the SCN produces a perturbation $\mathbf{\Delta}$ with $\|\mathbf{\Delta}\|_2 \leq \epsilon$, then the spectral gap deviation is bounded by

$$|g(\mathbf{A} + \mathbf{\Delta}) - g(\mathbf{A})| \leq 2\epsilon.$$

Proof. By Weyl's inequality, each eigenvalue shifts by at most $\|\mathbf{\Delta}\|_2$ in either direction. The worst-case gap change occurs when adjacent eigenvalues shift in opposite directions, giving the bound 2ϵ . $\square$

F Contraction Proofs for the WAC Monitor

F.1 Immediate Contraction Lemma

[Immediate Contraction] If $\|\mathcal{K}_t\|_{\text{op}} \leq 1 - \tau_t$ with $\tau_t > 0$, then for any $\mathbf{x} \in \mathcal{V}$,

$$\|\mathcal{K}_t \mathbf{x}\| \leq (1 - \tau_t) \|\mathbf{x}\|.$$

Hence the step map $\mathbf{x} \mapsto \mathcal{K}_t \mathbf{x}$ is a contraction with constant $1 - \tau_t$.

Proof. By definition of the operator norm,

$$\|\mathcal{K}_t \mathbf{x}\| \leq \|\mathcal{K}_t\|_{\text{op}} \|\mathbf{x}\| \leq (1 - \tau_t) \|\mathbf{x}\|.$$

Since $\tau_t > 0$, we have $1 - \tau_t < 1$, establishing the contraction property. $\square$

F.2 Time-Invariant Contraction

[Banach Fixed Point] Assume time-invariant weights so that $\mathcal{K}_t = \mathcal{K}$ and $\tau_t = \tau$ for some $\tau \in (0, 1)$. Then there exists a unique equilibrium $\mathbf{T}^* \in \mathcal{V}$ such that $\mathbf{T}^* = \mathcal{K}\mathbf{T}^* + \mathbf{F}$, and for any initial condition $\mathbf{T}_0$, the trajectory of the update map $\mathbf{U}$ satisfies

$$\|\mathbf{T}_t - \mathbf{T}^*\| \leq (1 - \tau)^t \|\mathbf{T}_0 - \mathbf{T}^*\|.$$

Proof. The update equation is $\mathbf{T}_{t+1} = \mathcal{K}\mathbf{T}_t + \mathbf{F}$. Define the affine map $\mathcal{A}(\mathbf{x}) = \mathcal{K}\mathbf{x} + \mathbf{F}$. For any two points $\mathbf{x}, \mathbf{y}$,

$$\|\mathcal{A}(\mathbf{x}) - \mathcal{A}(\mathbf{y})\| = \|\mathcal{K}(\mathbf{x} - \mathbf{y})\| \leq (1 - \tau) \|\mathbf{x} - \mathbf{y}\|.$$

By the Banach fixed-point theorem, $\mathcal{A}$ has a unique fixed point $\mathbf{T}^*$, and the iterates converge geometrically at rate $1 - \tau$. $\square$

F.3 Uniform Contraction Under Time-Varying Weights

[Distance Decay] Assume uniform gap $\tau_t \geq \tau > 0$ for all t . For any two trajectories $\{\mathbf{T}_t\}, \{\mathbf{T}'_t\}$ of $\mathbf{U}$ with initial conditions $\mathbf{T}_0, \mathbf{T}'_0$,

$$\|\mathbf{T}_t - \mathbf{T}'_t\| \leq (1 - \tau)^t \|\mathbf{T}_0 - \mathbf{T}'_0\|.$$

Proof. By induction. For $t = 0$, the claim is trivial. Assuming it holds for t ,

$$\|\mathbf{T}_{t+1} - \mathbf{T}'_{t+1}\| = \|\mathcal{K}_{t+1}(\mathbf{T}_t - \mathbf{T}'_t)\| \leq (1 - \tau) \|\mathbf{T}_t - \mathbf{T}'_t\| \leq (1 - \tau)^{t+1} \|\mathbf{T}_0 - \mathbf{T}'_0\|.$$

□

[Cauchy Limit] If $\sum_{t=0}^{\infty} \tau_t = \infty$ with $\tau_t \geq \tau > 0$, the sequence $\{\mathbf{T}_t\}$ is Cauchy and converges to a limit $\mathbf{T}_{\infty}$. If additionally $\mathcal{K}_t \rightarrow \mathcal{K}$ in operator norm, then $\mathbf{T}_{\infty}$ is the unique solution of $\mathbf{T} = \mathcal{K}\mathbf{T} + \mathbf{F}$.

Proof. Compose the contractive maps and use completeness of $\mathcal{V}$. The distance decay lemma shows that $\|\mathbf{T}_t - \mathbf{T}_s\| \rightarrow 0$ as $t, s \rightarrow \infty$, establishing the Cauchy property. Continuity of the fixed point under operator-norm convergence gives the final claim. □

G Windowed Average Contraction (WAC)

G.1 WAC Definition and Rate

[Windowed Average Contraction] Fix an integer window $m \geq 1$. Suppose there exists $\rho \in (0, 1)$ such that for all t ,

$$\prod_{j=t}^{t+m-1} \|\mathcal{K}_j\|_{\text{op}} \leq \rho.$$

We say the system satisfies the WAC condition with parameters (m, ρ) .

[Rate Under WAC] Under WAC with parameters (m, ρ) , for any $\mathbf{T}_0$,

$$\|\mathbf{T}_t - \mathbf{T}^*\| \leq C \cdot \rho^{\lfloor t/m \rfloor},$$

for some constant C depending on $\|\mathbf{T}_0 - \mathbf{T}^*\|$, m , and ρ .

Proof. Consider the m -step map $\mathcal{K}_t^{(m)} = \mathcal{K}_{t+m-1} \cdots \mathcal{K}_t$. By the WAC condition, $\|\mathcal{K}_t^{(m)}\|_{\text{op}} \leq \rho$. Apply Theorem 2.1 to the induced dynamics on blocks of length m . □

G.2 Telescoping WAC Product for ACE

[ACE WAC Monitor] Let $\bar{m}^{(t)}$ be the monitored scalar trajectory in the ACE runtime. Define the windowed contraction statistic

$$W_t = \frac{\bar{m}^{(t+\tau)}}{\bar{m}^{(t)}},$$

where $\tau = \tau_{\min}$ is the minimum window length. The ACE runtime certifies the trajectory as contractive if

$$\max_{t \in \mathcal{W}} W_t < 1.0,$$

where $\mathcal{W}$ is the set of all valid window starting positions in the holdout phase.

Proof. The ratio W_t measures the relative decay of the monitored quantity over a window of length τ . If $W_t < 1$ for all windows, the quantity is strictly decreasing on every window, implying global contraction over the holdout phase. □

H Drift with Sparse Expansions (DSE)

H.1 DSE Permissibility Condition

[Sparse Expansion] A step t is called an *expansion* if $\|\mathcal{K}_t\|_{\text{op}} \geq 1$. The system satisfies the DSE condition with parameters (M, ρ, δ) if within every macro-window of length M , the number of expansion steps is at most δM , and the product over the macro-window satisfies

$$\prod_{j=t}^{t+M-1} \|\mathcal{K}_j\|_{\text{op}} \leq \rho < 1.$$

[DSE Convergence] Under DSE with parameters (M, ρ, δ) , the conclusions of Theorem 3.3 hold with the modified rate $\rho^{1/M}$.

Proof. Partition the trajectory into blocks of length M . By the DSE condition, each block is contractive with factor ρ . Apply the WAC theorem with block size M and contraction factor ρ . $\square$

H.2 ACE DSE Tolerance Implementation

[ACE DSE Threshold] The ACE runtime implements a DSE tolerance with the following logic:

- **Strict mode:** $\max_t W_t < 1.0$ is required for certification.
- **DSE mode:** If $1.0 \leq \max_t W_t < 1.05$, the system applies a cognitive penalty but maintains certification.
- **Fail mode:** If $\max_t W_t \geq 1.05$, the system detects an unrecoverable arithmetic black hole and sets `certified = False`.

Justification. The threshold 1.05 is chosen such that the macro-window product over any horizon $M \geq 20$ remains contractive:

$$(1.05)^{\delta M} \cdot (1 - \tau)^{(1-\delta)M} < 1,$$

for $\delta \leq 0.05$ (5% drift tolerance) and $\tau \geq 0.1$ (minimum contraction gap). This ensures that sparse expansions cannot accumulate to break global contraction. $\square$

I Canonical Circuit Topology Invariant

I.1 Poseidon2 Parameter Lock

[Canonical Poseidon2 Topology] The ACE protocol requires a Poseidon2 sponge with strictly locked parameters:

$$t = 9, \quad r = 8,$$

where t is the total width and r is the rate (number of elements absorbed/squeezed per round).

[Constraint Budget Derivation] For the canonical topology ($t = 9, r = 8$) processing a 64-element vector, the R1CS constraint budget is:

$$\begin{aligned} \text{COST_FWHT} &= 384, \\ \text{COST_POSEIDON_H} &= 3171, \\ \text{COST_POSEIDON_GAMMA} &= 1500, \\ \text{COST_RANGE} &= 32. \end{aligned}$$

The total canonical budget is

$$B_{\text{canonical}} = 384 + 3171 + 1500 + 32 = 5087.$$

- Component Breakdown.*
- **FWHT (384)**: The Fast Walsh-Hadamard Transform on 64 elements requires $\log_2(64) = 6$ stages, each with $64/2 = 32$ butterfly operations. Each butterfly costs 2 constraints (addition and subtraction), giving $6 \times 32 \times 2 = 384$.
 - **Poseidon2 H (3171)**: The full permutation with $t = 9$ requires 8 full rounds and 22 partial rounds. Each full round applies the S-box to all 9 elements (9 constraints each) plus matrix multiplication ($9 \times 9 = 81$ constraints), giving $(8 + 22) \times (9 + 81) = 30 \times 90 = 2700$, plus additional constraints for constants and padding, totaling 3171.
 - **Poseidon2 Γ (1500)**: The gamma layer for domain separation and commitment binding adds 1500 constraints for the 64-element vector absorption.
 - **Range checks (32)**: Final range proofs for the 64 output elements, at 0.5 constraints per element on average.

□

I.2 Active Enforcement Invariant

[Circuit Violation Detection] Let C_{compiler} be the number of R1CS constraints generated by the upstream ZK compiler (Circom, Halo2, etc.). The ACE runtime enforces:

$$C_{\text{compiler}} = B_{\text{canonical}} = 5087.$$

If $C_{\text{compiler}} \neq 5087$, the runtime raises `CircuitViolationError` and fails closed.

Security Argument. A deviation in C_{compiler} indicates one of:

1. **Topology substitution**: The compiler used $t \neq 9$ or $r \neq 8$, altering the cryptographic properties.
2. **Algorithmic substitution**: The compiler replaced Poseidon2 with a different permutation (e.g., Rescue, Griffin), changing the security assumptions.
3. **Optimization error**: The compiler over-optimized, removing necessary constraints (e.g., missing range checks).
4. **Malicious injection**: The compiler added hidden constraints for backdoor channels.

All four cases break the formal correspondence between the ACE proof and the R8 patent claims. Active enforcement prevents silent substitution attacks. □

J Certificate Generation and Formal Properties

J.1 Certificate Structure

[ACECertificate] An ACE certificate is a tuple

$$\mathcal{C} = (\gamma, \theta, \sigma, \chi, \kappa, \omega, \beta, \iota),$$

where:

- γ : CAS commitment hash (derived from θ_{base} and nonce).
- θ : Base parameter block $(\epsilon, \epsilon_{\text{supp}}, \delta, N_0, K, M, \beta, \tau_{\text{min}}, \alpha_M)$.
- σ : Derived shuffle seed.
- χ : Certification status $\{\text{True}, \text{False}\}$.
- κ : Contraction intensity X_N .
- ω : Support size K .
- β : Maximum WAC product W_{max} .
- ι : Circuit budget and topology declaration.

J.2 Certificate Validity Conditions

[Certificate Validity] A certificate $\mathcal{C}$ is valid if and only if:

1. $\chi = \text{True}$ (explicit certification flag).
2. $\omega = K_{\text{governed}}$ (support size matches L1 governance output).
3. $\beta < 1.05$ (WAC/DSE threshold satisfied).
4. $\iota.\text{constraints} = 5087$ (circuit budget verified).
5. $\iota.\text{topology} = \text{"Poseidon2(t=9, r=8)"}$ (topology locked).

Proof. Conditions 1–2 ensure the governance layer was respected. Condition 3 ensures the recurrence monitor certified contraction (with DSE tolerance if applicable). Conditions 4–5 ensure the ZK backend was not substituted. The conjunction is necessary and sufficient because each condition guards a distinct attack surface. $\square$

K Summary of Guarantees

Guarantee	Formal Statement
Per-step contraction	$\ \mathcal{K}_t\ _{\text{op}} \leq 1 - \tau_t$ implies $\ \mathbf{T}_{t+1} - \mathbf{T}^*\ \leq (1 - \tau_t)\ \mathbf{T}_t - \mathbf{T}^*\ $
Global convergence (time-invariant)	$\ \mathbf{T}_t - \mathbf{T}^*\ \leq (1 - \tau)^t \ \mathbf{T}_0 - \mathbf{T}^*\ $
WAC rate	$\ \mathbf{T}_t - \mathbf{T}^*\ \leq C \cdot \rho^{\lfloor t/m \rfloor}$ under (m, ρ) -WAC
DSE rate	$\ \mathbf{T}_t - \mathbf{T}^*\ \leq C \cdot \rho^{\lfloor t/M \rfloor}$ under (M, ρ, δ) -DSE
Gap control	$ g(\mathbf{A} + \mathbf{\Delta}) - g(\mathbf{A}) \leq 2\ \mathbf{\Delta}\ _2$
Circuit lock	$C_{\text{compiler}} = 5087$ enforced at runtime
Certificate validity	Conjunction of 5 independent conditions

L Conclusion

This Mathematical Appendix has provided the complete formal foundation for the ACE certification runtime. All operator norm bounds, contraction proofs, and circuit invariants are derived from first principles and are parameterized by the specific ACE scaffold values. The DSE tolerance mechanism provides a principled bridge between strict mathematical purity and realistic biological noise, while the active circuit enforcement prevents silent substitution attacks on the zero-knowledge backend.

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